

Higher Tier only questions

| Question |     |  |  | Marking details                                                                                     | Marks Available                     |     |     |       |       |      |   |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------|-------------------------------------|-----|-----|-------|-------|------|---|
|          |     |  |  |                                                                                                     | AO1                                 | AO2 | AO3 | Total | Maths | Prac |   |
| 3        | (a) |  |  | Samples A, B and C contain magnesium or calcium ions                                                | <input checked="" type="checkbox"/> |     |     | 1     | 1     |      |   |
|          |     |  |  | Only samples A and B contain magnesium or calcium ions                                              | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | Only samples B and C contain magnesium or calcium ions                                              | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | None of the samples contain magnesium or calcium ions                                               | <input type="checkbox"/>            |     |     |       |       |      |   |
|          | (b) |  |  | To show the expected result for permanent hard water                                                | <input type="checkbox"/>            | 1   |     | 1     |       | 1    |   |
|          |     |  |  | To show the expected result for temporary hard water                                                | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | To act as a control for the experiment                                                              | <input checked="" type="checkbox"/> |     |     |       |       |      |   |
|          |     |  |  | To make sure the investigation is a fair test                                                       | <input type="checkbox"/>            |     |     |       |       |      |   |
|          | (c) |  |  | volume of soap solution and number of times shaken                                                  | <input type="checkbox"/>            | 1   |     | 1     |       | 1    |   |
|          |     |  |  | volume of water sample and volume of soap solution used                                             | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | volume of water sample and number of times shaken                                                   | <input checked="" type="checkbox"/> |     |     |       |       |      |   |
|          |     |  |  | volume of water sample, volume of soap solution and number of times shaken                          | <input type="checkbox"/>            |     |     |       |       |      |   |
|          | (d) |  |  | Sample B contains temporary hardness only                                                           | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | Sample B contains permanent hardness only                                                           | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | Sample B contains a mixture of temporary and permanent hardness                                     | <input checked="" type="checkbox"/> |     |     |       |       |      |   |
|          |     |  |  | Sample B does not contain hardness                                                                  | <input type="checkbox"/>            |     |     |       |       |      |   |
|          |     |  |  | the volume of soap needed after boiling is less than before / less soap is needed after boiling (1) |                                     |     |     |       |       |      |   |
|          |     |  |  | the volume of soap after boiling is still more than the distilled water (1)                         |                                     |     |     | 3     | 3     |      | 3 |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks Available |     |     |       |       |      |
|----------|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (e) |  |  | <p>if answered yes<br/>accept any two of following for (1) each</p> <ul style="list-style-type: none"><li>• strengthens bones / teeth</li><li>• reduced chance of heart disease</li><li>• better for baking / brewing industry</li><li>• improves flavour of water</li></ul> <p>if answered no<br/>accept any two of following for (1) each</p> <ul style="list-style-type: none"><li>• forms a scum with soap / wastes soap</li><li>• furs up kettles / pipes / forms limescale</li><li>• appliances become less efficient (or named appliance)</li><li>• bad taste / tastes different</li></ul> <p>if answered 'unable to decide' must have one reason in favour and one against to gain (2)</p> <p>if no opinion is stated, credit may still be awarded if the reasons clearly imply an opinion</p> <p>neutral: reference to cost / calcium / magnesium<br/>accept other creditworthy answers</p> | 2               |     |     | 2     |       |      |
|          |     |  |  | Question 3 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4               | 0   | 4   | 8     | 0     | 5    |